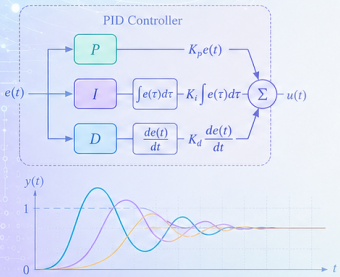
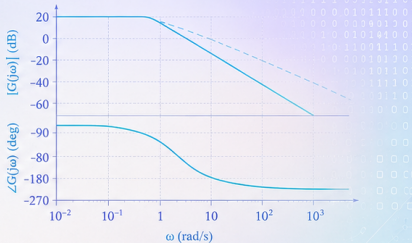
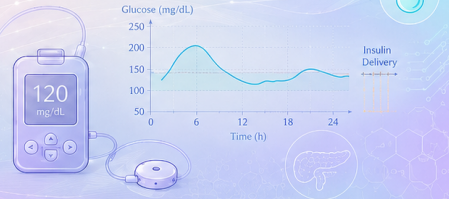
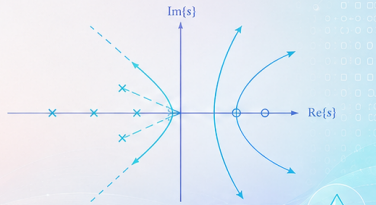


Frequency Response

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$$G(s) = \frac{Y(s)}{U(s)} \quad a_n \frac{d^n y(t)}{dt^n} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \dots + b_m \frac{d^m u(t)}{dt^m} + \dots + b_1 \frac{du(t)}{dt} + b_0 u(t)$$



Frequency Response

Consider an LTI system

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + B\mathbf{u}(t),$$

$$\mathbf{y}(t) = C\mathbf{x}(t) + D\mathbf{u}(t).$$

Its transfer function is

$$G(s) = C(sI - A)^{-1}B + D$$

- The **frequency response** describes the response to a sinusoidal input.

$$u(t) = \sin(\omega t + \phi)$$

Frequency Response

For a stable LTI system, the steady-state output is

$$y_{ss}(t) = |G(j\omega)| \sin[\omega t + \phi + \angle G(j\omega)]$$

Therefore,

$$|G(j\omega)|$$

determines the amplitude scaling, while

$$\angle G(j\omega)$$

determines the phase shift.

Physical Interpretation of Frequency Response

Suppose the sinusoidal input is

$$u(t) = A_u \sin(\omega t).$$

Then, after the transient response disappears,

$$y_{ss}(t) = A_y \sin(\omega t + \theta).$$

The **magnitude** is obtained from the amplitude ratio:

$$|G(j\omega)| = \frac{A_y}{A_u}$$

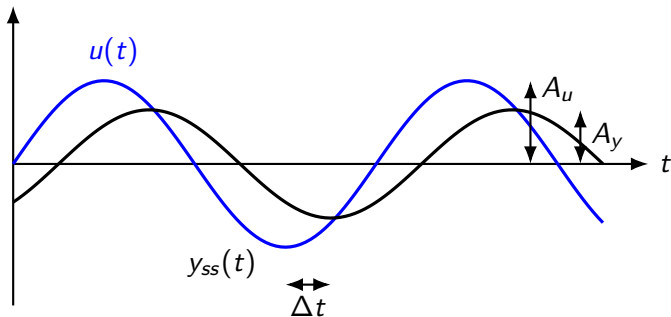
If $|G(j\omega)| > 1$, the sinusoid is amplified.

If $|G(j\omega)| < 1$, the sinusoid is attenuated.

The **phase** is determined from the relative time shift. If the output lags the input by Δt , $\angle G(j\omega) = -\omega \Delta t$ where the phase is in radians.

Thus, frequency response describes **Gain and phase versus frequency**

Sinusoidal Input and Output



$$|G(j\omega)| = \frac{A_y}{A_u}, \quad \angle G(j\omega) = -\omega \Delta t$$

Important

The input and steady-state output have the **same frequency**; only their amplitude and phase are changed.

Why Does the Output Have the Same Frequency?

Using Euler's formula,

$$\sin(\omega t + \phi) = \frac{e^{j(\omega t + \phi)} - e^{-j(\omega t + \phi)}}{2j}.$$

For a real LTI system,

$$G(-j\omega) = G^*(j\omega).$$

Write

$$G(j\omega) = |G(j\omega)|e^{j\angle G(j\omega)}.$$

Hence,

$$G(-j\omega) = |G(j\omega)|e^{-j\angle G(j\omega)}.$$

Key Property

A complex exponential $e^{j\omega t}$ passing through an LTI system is multiplied by $G(j\omega)$, but its frequency remains ω .

Frequency Response: Derivation

By superposition, the steady-state response is

$$y_{ss}(t) = \frac{1}{2j} G(j\omega) e^{j(\omega t + \phi)} - \frac{1}{2j} G(-j\omega) e^{-j(\omega t + \phi)}.$$

Substituting

$$G(j\omega) = |G(j\omega)| e^{j\angle G(j\omega)},$$

and

$$G(-j\omega) = |G(j\omega)| e^{-j\angle G(j\omega)},$$

gives

$$y_{ss}(t) = |G(j\omega)| \sin [\omega t + \phi + \angle G(j\omega)]$$

Thus, the frequency response is completely described by

$$|G(j\omega)|, \quad \angle G(j\omega)$$

Empirical Determination of Frequency Response

The frequency response can also be determined **experimentally**, even when the mathematical model is unknown.

1. Apply a sinusoidal input at a fixed frequency

$$u(t) = A_u \sin(\omega t).$$

2. Wait until the transient response disappears.
3. Measure the output amplitude A_y and time shift Δt .

Calculate

$$|G(j\omega)| = \frac{A_y}{A_u},$$

and

$$\angle G(j\omega) = -\omega \Delta t.$$

Then repeat the experiment for different values of ω .

Result

The measured magnitude and phase values provide an empirical **frequency-response model**.

Frequency-Domain Plots

Plotting $G(j\omega)$ as the frequency varies provides useful insight into the behaviour of an LTI system.

Bode Plot

Magnitude:

$$20 \log_{10} |G(j\omega)|$$

and phase:

$$\angle G(j\omega)$$

plotted versus logarithmic frequency.

Polar Plot

Plot $\text{Im } G(j\omega)$ against $\text{Re } G(j\omega)$.

Magnitude–Phase Plot

Plot

$$20 \log_{10} |G(j\omega)|$$

against

$$\angle G(j\omega).$$

These representations are useful for studying

- ▶ frequency-selective behaviour,
- ▶ bandwidth,
- ▶ resonant behaviour,
- ▶ and stability margins.

Decibel Units

The **Bel** represents a logarithmic power ratio:

$$\log_{10} \left(\frac{P}{P_0} \right) \text{ Bel.}$$

The more commonly used **decibel** is

$$10 \log_{10} \left(\frac{P}{P_0} \right) \text{ dB.}$$

For an LTI system,

$$Y(j\omega) = G(j\omega)U(j\omega).$$

Hence,

$$|Y(j\omega)|^2 = |G(j\omega)|^2 |U(j\omega)|^2.$$

Therefore, the output-to-input power ratio is

$$10 \log_{10} |G(j\omega)|^2.$$

Thus,

$$\boxed{20 \log_{10} |G(j\omega)| \text{ dB}}$$

is used for transfer-function magnitude.

Bode Plot

A **Bode plot** represents the frequency response using two plots:

Magnitude Plot

$$20 \log_{10} |G(j\omega)|$$

in decibels versus

$$\log_{10} \omega.$$

Phase Plot

$$\angle G(j\omega)$$

in degrees versus

$$\log_{10} \omega.$$

The logarithmic frequency scale allows a very wide range of frequencies to be displayed compactly.

$$\omega : 10^{-2}, 10^{-1}, 10^0, 10^1, 10^2, \dots$$

Decades and Octaves

A logarithmic frequency scale is divided into **decades** and **octaves**.

The number of decades between ω_1 and ω_2 is

$$N_{\text{dec}} = \log_{10} \left(\frac{\omega_2}{\omega_1} \right)$$

One decade corresponds to

$$\omega_2 = 10\omega_1.$$

Therefore,

$$20 \text{ dB/decade} \approx 6 \text{ dB/octave.}$$

The number of octaves is

$$N_{\text{oct}} = \log_2 \left(\frac{\omega_2}{\omega_1} \right)$$

One octave corresponds to

$$\omega_2 = 2\omega_1.$$

Also,

$$1 \text{ decade} \approx 3.32 \text{ octaves.}$$

Magnitude and Phase of a Transfer Function

Consider

$$G(s) = \frac{b_1(s)b_2(s)}{a_1(s)a_2(s)}.$$

Magnitude

$$|G(s)| = \frac{|b_1(s)||b_2(s)|}{|a_1(s)||a_2(s)|}.$$

Phase

$$\begin{aligned}\angle G(s) &= \angle b_1(s) + \angle b_2(s) \\ &\quad - \angle a_1(s) - \angle a_2(s).\end{aligned}$$

Taking logarithms,

$$\begin{aligned}\log |G(s)| &= \log |b_1(s)| + \log |b_2(s)| \\ &\quad - \log |a_1(s)| - \log |a_2(s)|.\end{aligned}$$

Thus, individual pole and zero contributions can be **added graphically**.

Key Advantage of Bode Plots

Multiplication and division of transfer-function factors become addition and subtraction of their Bode plots.

Transfer Function in Bode Form

A general transfer function may be decomposed into elementary factors:

$$G(s) = \kappa \frac{\prod_i \left(1 + \frac{s}{z_i}\right) \prod_l \left[\left(\frac{s}{\omega_{nl}}\right)^2 + 2\zeta_l \frac{s}{\omega_{nl}} + 1\right]}{s^{n_0} \prod_i \left(1 + \frac{s}{p_i}\right) \prod_k \left[\left(\frac{s}{\omega_{nk}}\right)^2 + 2\zeta_k \frac{s}{\omega_{nk}} + 1\right]}.$$

The four basic factors are

- ▶ constant gain κ ,
- ▶ poles or zeros at the origin,
- ▶ real poles or zeros,
- ▶ complex-conjugate pole or zero pairs.

Once the magnitude and phase contributions of these basic factors are known, the Bode plot of a complicated transfer function is obtained by

adding their individual contributions.

Bode Plot of a Constant Gain

Consider

$$G(s) = \kappa.$$

Magnitude

$$20 \log_{10} |\kappa|$$

The magnitude is independent of frequency.

Examples:

$$\kappa = 10 \quad \Rightarrow \quad 20 \text{ dB},$$

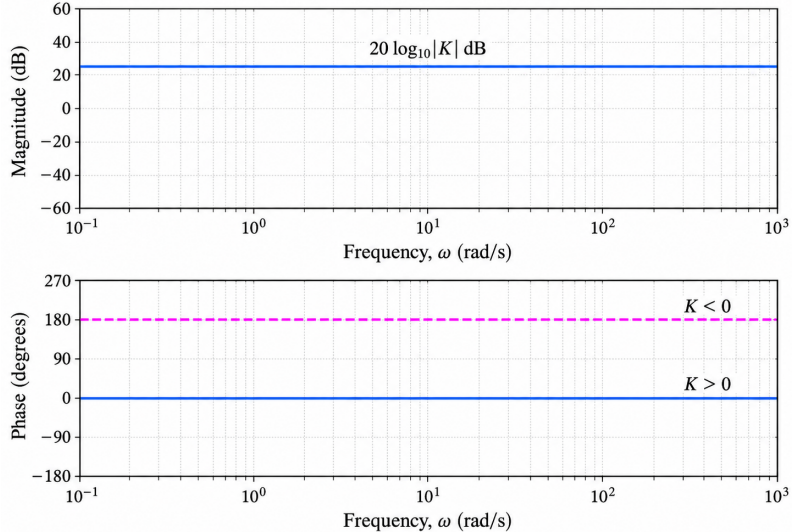
$$\kappa = 0.1 \quad \Rightarrow \quad -20 \text{ dB}.$$

Phase

$$\angle \kappa = \begin{cases} 0^\circ, & \kappa > 0, \\ 180^\circ, & \kappa < 0. \end{cases}$$

Thus both magnitude and phase are **horizontal lines** in the Bode plot.

Bode Plot of a Constant Gain



Pole or Zero at the Origin

Consider

$$G(s) = s^q.$$

For $s = j\omega$,

$$G(j\omega) = (j\omega)^q.$$

Magnitude

$$20 \log_{10} |(j\omega)^q| = 20q \log_{10} \omega.$$

Thus, the slope is

$$20q \text{ dB/decade}$$

and the line passes through 0 dB at

$$\omega = 1.$$

Phase

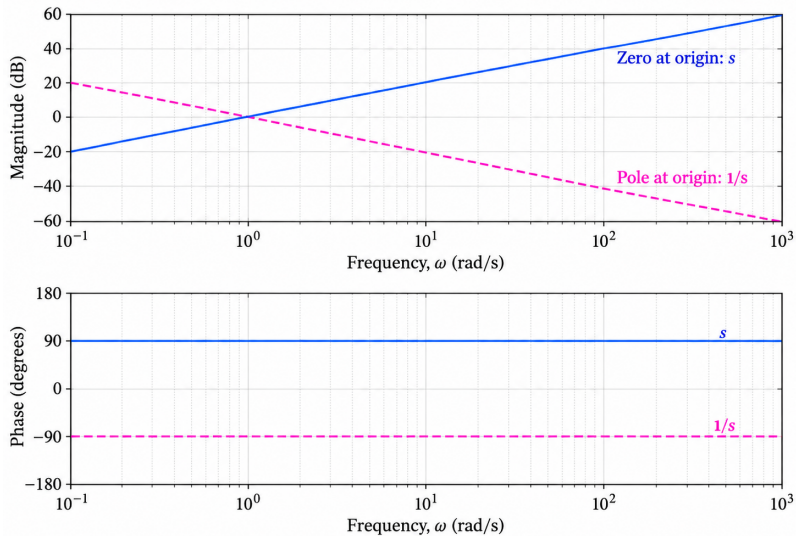
$$\angle(j\omega)^q = 90q^\circ$$

Examples:

$$s : \quad +20 \text{ dB/dec}, \quad +90^\circ,$$

$$\frac{1}{s} : \quad -20 \text{ dB/dec}, \quad -90^\circ.$$

Bode Plot of a Pole/Zero at the Origin



Bode Plot of a Real Zero

Consider a real zero

$$G(s) = 1 + \frac{s}{z}.$$

At $s = j\omega$,

$$G(j\omega) = 1 + j\frac{\omega}{z}.$$

Magnitude

$$|G(j\omega)| = \sqrt{1 + \left(\frac{\omega}{z}\right)^2}.$$

Hence,

$$20 \log_{10} |G(j\omega)| = 10 \log_{10} \left[1 + \left(\frac{\omega}{z}\right)^2 \right]$$

Phase

$$\angle G(j\omega) = \tan^{-1} \left(\frac{\omega}{z} \right)$$

The frequency $\omega = z$ is called the **corner frequency** or break frequency.

Real Zero: Asymptotic Approximation

For

$$G(s) = 1 + \frac{s}{z},$$

three frequency regions are important.

Low Frequency: $\omega \ll z$

$$|G(j\omega)| \approx 1$$

$$20 \log_{10} |G(j\omega)| \approx 0 \text{ dB}$$

and

$$\angle G(j\omega) \approx 0^\circ.$$

At $\omega = z$

$$|G(jz)| = \sqrt{2}$$

$$20 \log_{10} \sqrt{2} \approx 3 \text{ dB and } \angle G(jz) = 45^\circ$$

High Frequency: $\omega \gg z$

$$G(j\omega) \approx j \frac{\omega}{z}.$$

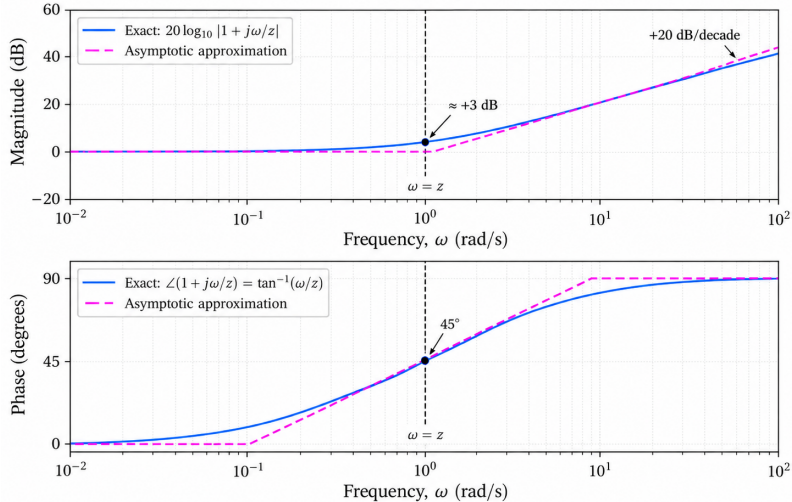
Therefore,

$$20 \log_{10} |G(j\omega)| \approx 20 \log_{10} \left(\frac{\omega}{z} \right).$$

Thus, after $\omega = z$,
slope = +20 dB/decade and
 $\angle G(j\omega) \rightarrow 90^\circ$.

Bode Plot of a Real Zero

$$G(s) = 1 + \frac{s}{z}$$



Bode Plot of a Real Pole

Consider a real pole

$$G(s) = \frac{1}{1 + \frac{s}{p}}.$$

At $s = j\omega$,

$$G(j\omega) = \frac{1}{1 + j\frac{\omega}{p}}.$$

Magnitude

$$|G(j\omega)| = \frac{1}{\sqrt{1 + \left(\frac{\omega}{p}\right)^2}}.$$

Phase

$$\angle G(j\omega) = -\tan^{-1} \left(\frac{\omega}{p} \right)$$

Hence,

$$|20 \log_{10} |G(j\omega)| = -10 \log_{10} \left[1 + \left(\frac{\omega}{p} \right)^2 \right]$$

The frequency $\boxed{\omega = p}$ is the **corner frequency**.

Real Pole: Asymptotic Approximation

For

$$G(s) = \frac{1}{1 + \frac{s}{p}},$$

Low Frequency: $\omega \ll p$

$$|G(j\omega)| \approx 1$$

$$20 \log_{10} |G(j\omega)| \approx 0 \text{ dB}$$

and

$$\angle G(j\omega) \approx 0^\circ.$$

At the corner frequency, $\omega = p$,

$$20 \log_{10} |G(jp)| \approx -3 \text{ dB}$$

$$\text{and } \angle G(jp) = -45^\circ$$

High Frequency: $\omega \gg p$

$$G(j\omega) \approx \frac{p}{j\omega}.$$

Therefore, slope = -20 dB/decade and

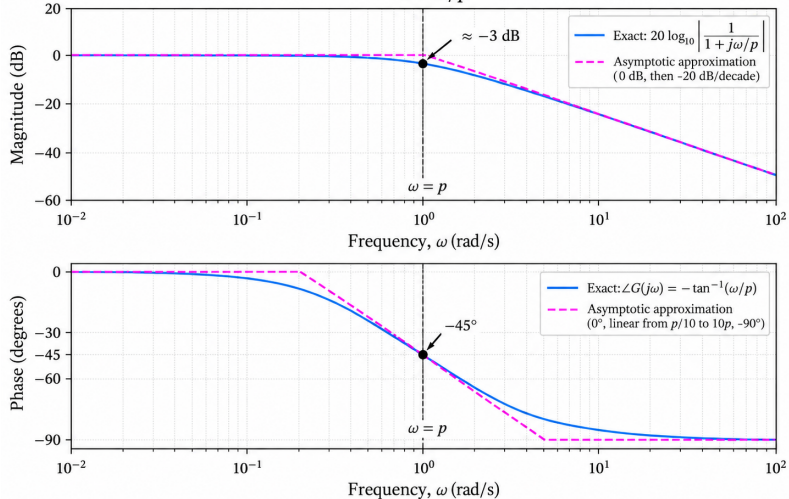
$$\angle G(j\omega) \rightarrow -90^\circ.$$

Interpretation

A real pole behaves approximately like a **constant at low frequency** and like an **integrator at high frequency**.

Bode Plot of a Real Pole

$$G(s) = \frac{1}{1 + s/p}$$



Bode Plot Example 1

Draw the Bode plot for

$$G(s) = 10 \frac{s + 10}{(s + 1)(s + 100)}.$$

Write the transfer function in Bode form:

$$G(s) = \frac{\left(1 + \frac{s}{10}\right)}{(1 + s) \left(1 + \frac{s}{100}\right)}$$

Step 1: Corner frequencies

$$\omega = 1, \quad 10, \quad 100$$

corresponding to

- ▶ pole at $\omega = 1$,
- ▶ zero at $\omega = 10$,
- ▶ pole at $\omega = 100$.

Step 2: Starting value At low frequency,

$$G(0) = 1.$$

Therefore,

$$20 \log_{10} |G(0)| = 0 \text{ dB}$$

and

$$\angle G(0) = 0^\circ.$$

Bode Plot Example 1: Magnitude Slopes

$$G(s) = \frac{\left(1 + \frac{s}{10}\right)}{(1 + s) \left(1 + \frac{s}{100}\right)}$$

Frequency Range	Active Factors	Net Slope
$\omega < 1$	None	0 dB/dec
$1 < \omega < 10$	Pole at 1	-20 dB/dec
$10 < \omega < 100$	Pole + Zero	0 dB/dec
$\omega > 100$	Two poles + one zero	-20 dB/dec

Rule

Each real pole changes the slope by **-20 dB/decade**, while each real zero changes it by **+20 dB/decade**.

Bode Plot Example 1: Phase Approximation

The phase is

$$\angle G(j\omega) = \tan^{-1}\left(\frac{\omega}{10}\right) - \tan^{-1}(\omega) - \tan^{-1}\left(\frac{\omega}{100}\right).$$

A real pole at $\omega = p$ contributes approximately

$$0^\circ \quad \text{for } \omega < 0.1p,$$

$$-45^\circ \quad \text{at } \omega = p,$$

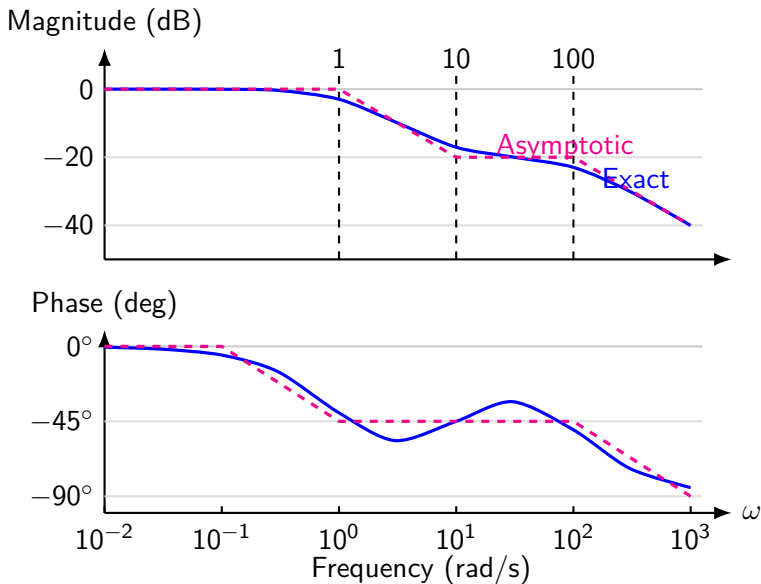
$$-90^\circ \quad \text{for } \omega > 10p.$$

A real zero at $\omega = z$ contributes approximately

$$0^\circ \quad \text{for } \omega < 0.1z,$$

$$+45^\circ \quad \text{at } \omega = z,$$

$$+90^\circ \quad \text{for } \omega > 10z.$$



Complex-Conjugate Zeros

Consider the second-order zero factor

$$G(s) = \left(\frac{s}{\omega_n} \right)^2 + 2\zeta \left(\frac{s}{\omega_n} \right) + 1.$$

Let

$$u = \frac{\omega}{\omega_n}.$$

Then

$$G(j\omega) = 1 - u^2 + j2\zeta u.$$

Magnitude

$$|G(j\omega)| = \sqrt{(1 - u^2)^2 + 4\zeta^2 u^2}$$

Thus,

$$20 \log_{10} |G(j\omega)|$$

gives the magnitude contribution.

Phase

$$\angle G(j\omega) = \tan^{-1} \left(\frac{2\zeta u}{1 - u^2} \right)$$

with appropriate quadrant correction.

At

$$\omega = \omega_n,$$

$$\angle G(j\omega_n) = 90^\circ.$$

Complex Zeros: Asymptotic Behaviour

For a complex-conjugate zero pair,

$$G(s) = \left(\frac{s}{\omega_n} \right)^2 + 2\zeta \frac{s}{\omega_n} + 1.$$

Low frequency For $\omega \ll \omega_n$,

$$|G(j\omega)| \approx 1,$$

so

$$20 \log_{10} |G(j\omega)| \approx 0 \text{ dB}.$$

Also,

$$\angle G(j\omega) \approx 0^\circ.$$

At $\omega = \omega_n$,

$$20 \log_{10} |G(j\omega_n)| = 20 \log_{10}(2\zeta)$$

High frequency For $\omega \gg \omega_n$,

$$|G(j\omega)| \approx \left(\frac{\omega}{\omega_n} \right)^2.$$

Thus,

slope = +40 dB/decade

and

$$\angle G(j\omega) \rightarrow 180^\circ.$$

Complex-Conjugate Poles

Consider

$$G(s) = \frac{1}{\left(\frac{s}{\omega_n}\right)^2 + 2\zeta\frac{s}{\omega_n} + 1}.$$

Let $u = \frac{\omega}{\omega_n}$.

Magnitude

$$|G(j\omega)| = \frac{1}{\sqrt{(1 - u^2)^2 + 4\zeta^2 u^2}}$$

At $\omega = \omega_n$,

$$|G(j\omega_n)| = \frac{1}{2\zeta}$$

Phase

$$\angle G(j\omega) = -\tan^{-1} \left(\frac{2\zeta u}{1 - u^2} \right)$$

with quadrant correction. At $\omega = \omega_n$,

$$\angle G(j\omega_n) = -90^\circ$$

Complex Poles: Asymptotic Behaviour

For

$$\omega \ll \omega_n,$$

the complex pole pair behaves approximately as a constant:

$$|G(j\omega)| \approx 1,$$

$$20 \log_{10} |G(j\omega)| \approx 0 \text{ dB},$$

$$\angle G(j\omega) \approx 0^\circ.$$

For

$$\omega \gg \omega_n,$$

$$|G(j\omega)| \approx \left(\frac{\omega_n}{\omega}\right)^2.$$

Therefore,

$\text{slope} = -40 \text{ dB/decade}$

and

$\angle G(j\omega) \rightarrow -180^\circ.$

Corner Frequency

The natural frequency $\omega = \omega_n$ acts as the break frequency of the complex pole pair.

Resonance and Damping Ratio

For

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2},$$

the magnitude is

$$|G(j\omega)| = \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + 4\zeta^2 \frac{\omega^2}{\omega_n^2}}}.$$

For

$$\zeta < \frac{1}{\sqrt{2}},$$

a resonant peak occurs at

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2}$$

The resonant peak is

$$M_r = \frac{1}{2\zeta \sqrt{1 - \zeta^2}}$$

For small ζ , $\omega_r \approx \omega_n$, $M_r \approx \frac{1}{2\zeta}$.

Effect of Damping on Frequency Response

- ▶ Small ζ produces a pronounced **resonant peak** near ω_n .
- ▶ Increasing ζ reduces the resonant peak.
- ▶ When

$$\zeta \geq \frac{1}{\sqrt{2}},$$

no resonant peak occurs.

- ▶ The asymptotic magnitude approximation is least accurate near

$$\boxed{\omega = \omega_n}.$$

- ▶ Both the exact magnitude and phase near ω_n depend strongly on the damping ratio.

Summary of Basic Bode-Plot Factors

Factor	Magnitude Slope	Phase Change
K	0	0° for $K > 0$
s	+20 dB/dec	$+90^\circ$
$\frac{1}{s}$	-20 dB/dec	-90°
$1 + \frac{s}{z}$	+20 dB/dec	$0^\circ \rightarrow 90^\circ$
$\frac{1}{1 + s/p}$	-20 dB/dec	$0^\circ \rightarrow -90^\circ$
Complex zero pair	+40 dB/dec	$0^\circ \rightarrow 180^\circ$
Complex pole pair	-40 dB/dec	$0^\circ \rightarrow -180^\circ$

LTI Systems as Filters

A frequency response allows an LTI system to be interpreted as a **filter**.

Low-Pass

$$G_{LP}(s) = \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$$

Passes low frequencies and attenuates high frequencies.

Band-Pass

$$G_{BP}(s) = \frac{2\zeta\omega_0 s}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$$

Passes a band of frequencies near ω_0 .

High-Pass

$$G_{HP}(s) = \frac{s^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$$

Passes high frequencies and attenuates low frequencies.

Frequency-Selective Behaviour of Filters

Filter	Low ω	Near ω_0	High ω
Low-pass	Pass	Transition	Attenuate
Band-pass	Attenuate	Pass	Attenuate
High-pass	Attenuate	Transition	Pass

Interpretation

The magnitude

$$|G(j\omega)|$$

indicates which frequency components of the input are **passed**, **amplified**, or **attenuated**.

Bode Plot Example 2

Consider

$$G(s) = \frac{k(s + b)}{(s + a)(s^2 + 2\zeta\omega_0 s + \omega_0^2)},$$

with $a \ll b \ll \omega_0$.

At low frequency,

$$G(0) = \frac{kb}{a\omega_0^2}.$$

Hence the initial magnitude is

$$20 \log_{10} \left| \frac{kb}{a\omega_0^2} \right|$$

The break frequencies are

$$a, \quad b, \quad \omega_0.$$

Slope changes:

$$\omega = a : \quad -20 \text{ dB/dec},$$

$$\omega = b : \quad +20 \text{ dB/dec},$$

$$\omega = \omega_0 : \quad -40 \text{ dB/dec}.$$

Therefore the final high-frequency slope is

$$-40 \text{ dB/dec}.$$

Bode Plot Example 2: Phase Construction

For

$$G(s) = \frac{k(s + b)}{(s + a)(s^2 + 2\zeta\omega_0s + \omega_0^2)},$$

the phase contributions are

$$\angle G(j\omega) = \angle k + \angle(j\omega + b) - \angle(j\omega + a) - \angle G_2(j\omega).$$

- ▶ Real pole at a : approximately $0^\circ \rightarrow -90^\circ$.
- ▶ Real zero at b : approximately $0^\circ \rightarrow +90^\circ$.
- ▶ Complex pole pair at ω_0 : approximately $0^\circ \rightarrow -180^\circ$.

Approximation Rule

The phase transition of a real pole or zero is usually approximated from one decade below to one decade above its corner frequency.

Bode Plot Example 3

Consider

$$G(s) = \frac{4(1 + 0.1s)}{s(1 + 0.5s) \left[1 + 0.4 \left(\frac{s}{8} \right) + \left(\frac{s}{8} \right)^2 \right]}.$$

The factors, in increasing order of frequency, are

1. constant gain: $K = 4$;
2. pole at the origin:
3. real pole: $\omega = 2$;
4. real zero: $\omega = 10$;
5. complex pole pair: $\omega_n = 8$.

$$\frac{1}{s};$$

Example 3: Individual Magnitude Contributions

For

$$G(s) = \frac{4(1 + 0.1s)}{s(1 + 0.5s)[1 + 0.4(s/8) + (s/8)^2]},$$

Constant gain

$$20 \log_{10} 4 \approx 12.04 \text{ dB.}$$

Pole at origin

$$-20 \text{ dB/decade.}$$

Real pole at 2 Adds

$$-20 \text{ dB/decade}$$

after $\omega = 2$.

Zero at 10 Adds

$$+20 \text{ dB/decade}$$

after $\omega = 10$. Complex poles at 8 Add

$$-40 \text{ dB/decade}$$

after approximately $\omega = 8$.

Example 3: Net Magnitude Slope

The slope changes whenever a corner frequency is crossed.

Frequency Range	Net Slope
$\omega < 2$	-20 dB/dec
$2 < \omega < 8$	-40 dB/dec
$8 < \omega < 10$	-80 dB/dec
$\omega > 10$	-60 dB/dec

Important

At each corner frequency, add the slope contribution of the new pole or zero to the slope already present.

Example 3: Phase Contributions

The approximate phase contributions are

Factor	Phase Contribution
$K = 4$	0°
Pole at origin	-90°
Pole at $\omega = 2$	$0^\circ \rightarrow -90^\circ$
Zero at $\omega = 10$	$0^\circ \rightarrow +90^\circ$
Complex poles at $\omega_n = 8$	$0^\circ \rightarrow -180^\circ$

The total phase is obtained by

$$\angle G(j\omega) = \sum \text{individual phase contributions.}$$

Bode Plot Example 4

Consider

$$G(s) = \frac{(s + 1)(s^2 + 3s + 100)}{s^2(s + 10)(s + 100)}.$$

Writing it in Bode form,

$$G(s) = \frac{(s + 1) \left[\left(\frac{s}{10} \right)^2 + 2(0.15) \frac{s}{10} + 1 \right]}{10s^2 \left(1 + \frac{s}{10} \right) \left(1 + \frac{s}{100} \right)}$$

The important frequencies are

$$1, \quad 10, \quad 100 \text{ rad/s}$$

with a double pole at the origin.

Example 4: Factors and Their Effects

Factor	Corner Frequency	Magnitude Effect
Double pole at origin	0	-40 dB/dec
Real zero	1	+20 dB/dec
Complex zero pair	10	+40 dB/dec
Real pole	10	-20 dB/dec
Real pole	100	-20 dB/dec

At sufficiently low frequency,

$$G(j\omega) \approx \frac{1}{10(j\omega)^2}.$$

Hence,

$$\angle G(j\omega) \approx -180^\circ.$$

Example 4: Net Magnitude Slopes

Frequency Range	Net Magnitude Slope
$0.1 < \omega < 1$	-40 dB/dec
$1 < \omega < 10$	-20 dB/dec
$10 < \omega < 100$	0 dB/dec
$\omega > 100$	-20 dB/dec

At $\omega = 10$, two effects occur simultaneously:

$$\underbrace{+40}_{\text{complex zeros}} + \underbrace{(-20)}_{\text{real pole}} = \boxed{+20 \text{ dB/dec change}}.$$

Thank You

Questions and Discussion